Exploratory Data Analysis on the automobile.txt Dataset

Report

Introduction

The automobile.txt dataset contains a list of cars and records their: make, fuel type, aspiration, number of doors, body style, drive wheels, engine location, wheel base, length, width, engine size, fuel system, bore, stroke, compression ratio, horsepower, peak rpm, city miles per gallon (mpg), highway mpg and price.

In this EDA report, I will clean the data and extract information and visualisations about the various cars recorded in this dataset.

Data Cleaning

1. Dropping "Normalised Losses" and "Symboling":

For this EDA, "Normalised Losses" and "Symboling" are irrelevant… Because of this, these two columns have been removed from the dataset.

2. Removing Duplicate Rows:

Duplicate rows within the dataset can skew the data and cause inaccurate conclusions; therefore, they have been removed.

3. Remove Rows with Missing Data:

Some of the vehicles within this dataset have missing data, meaning that their values have not been recorded, or there is information that is missing. To avoid problems with these vehicles, they have been removed from the dataset and excluded from the findings of this EDA report. It is important to note that only data with non-numeric missing data was removed.

4. Converting all Numeric Column Cells to integers:

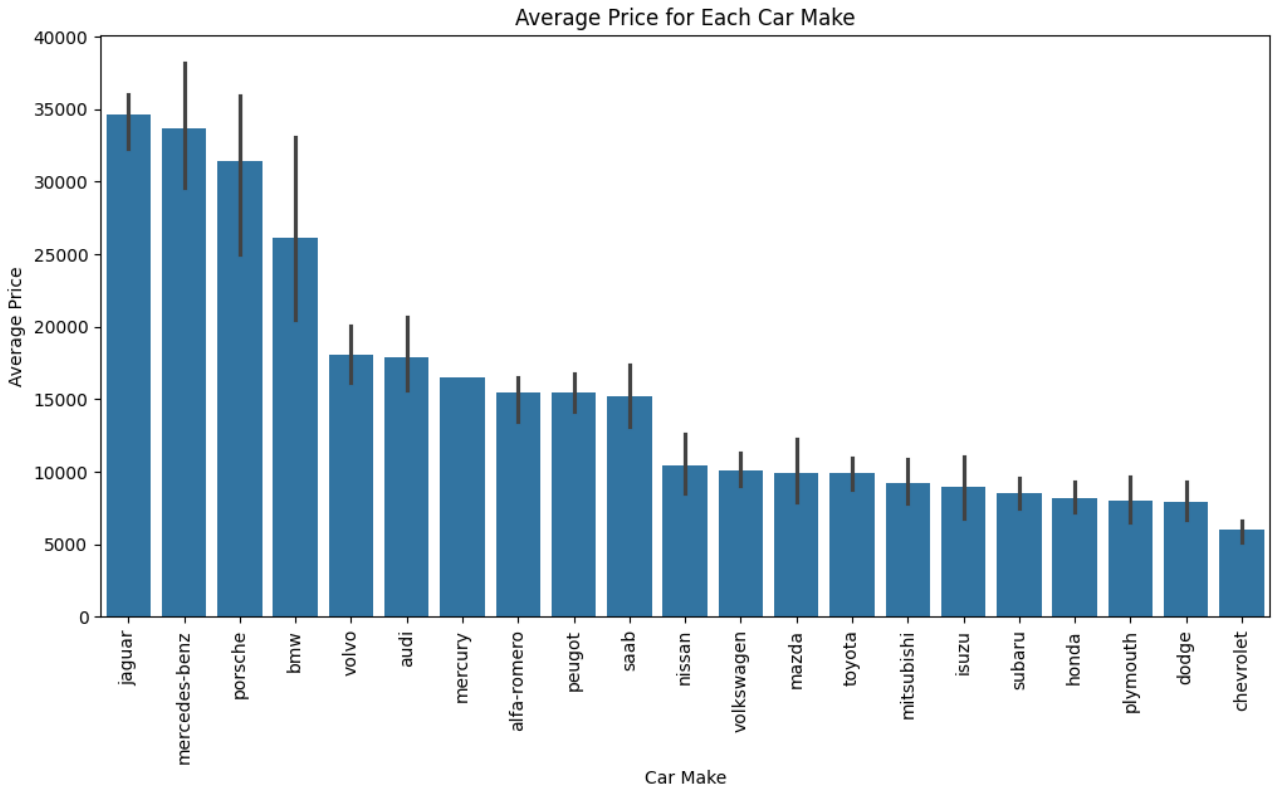
For the columns that are supposed to contain numerical data, some of them contained strings, e.g. "NaN" and "?". Therefore, they were all converted to integers using NumPy's 'int64' method in order to avoid skewed data representations later.

Data Stories and Visualisations

1. Most Expensive Cars:

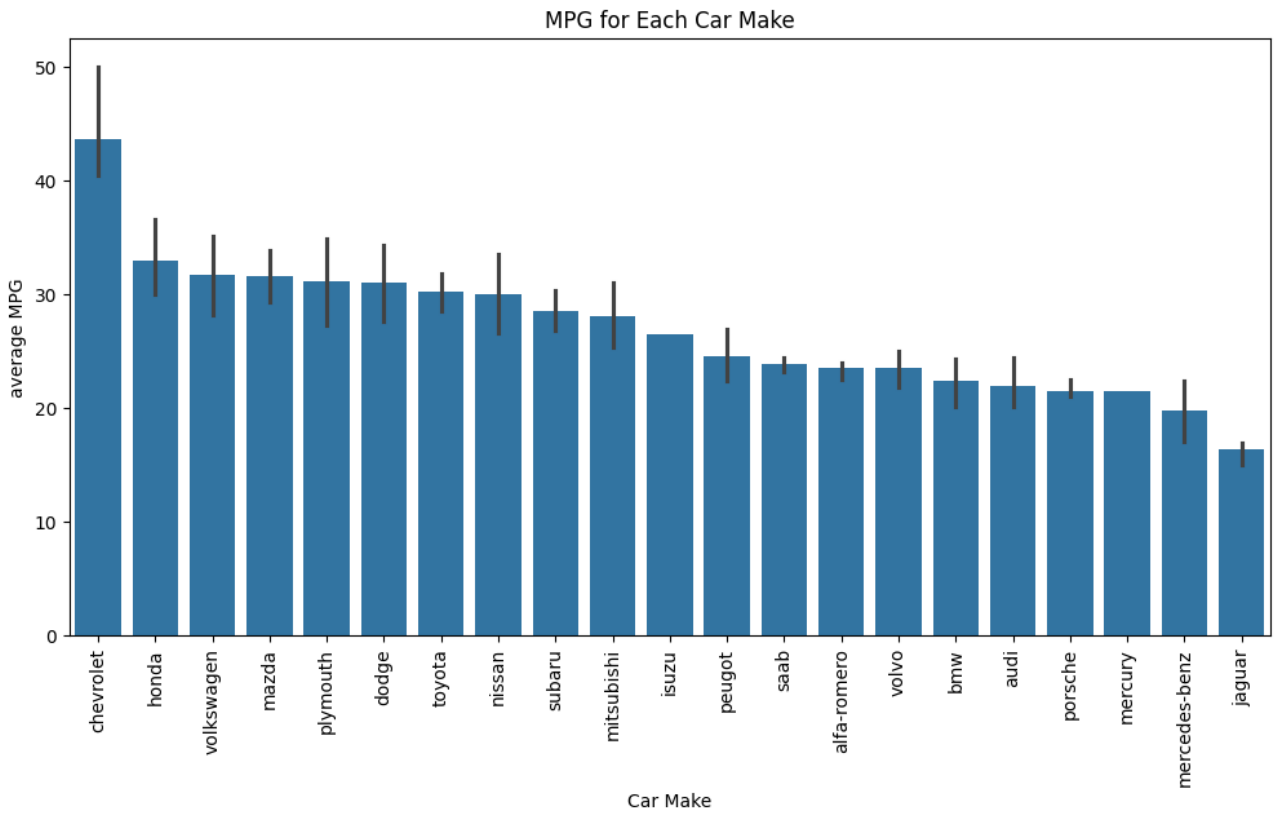
The top 5 most expensive cars are:
1. #74 Mercedes-Benz - 45 400
2. #16 BMW - 41 315
3. #73 Mercedes-Benz - 40 960
4. #128 Porsche - 37 028
5. #17 BMW - 36 880

Mercedes-Benz has the most expensive car, as well as 2 within the top 5, and BMW also has 2 within the top 5. Please note that the dataset does not tell us what currency the price was recorded in.



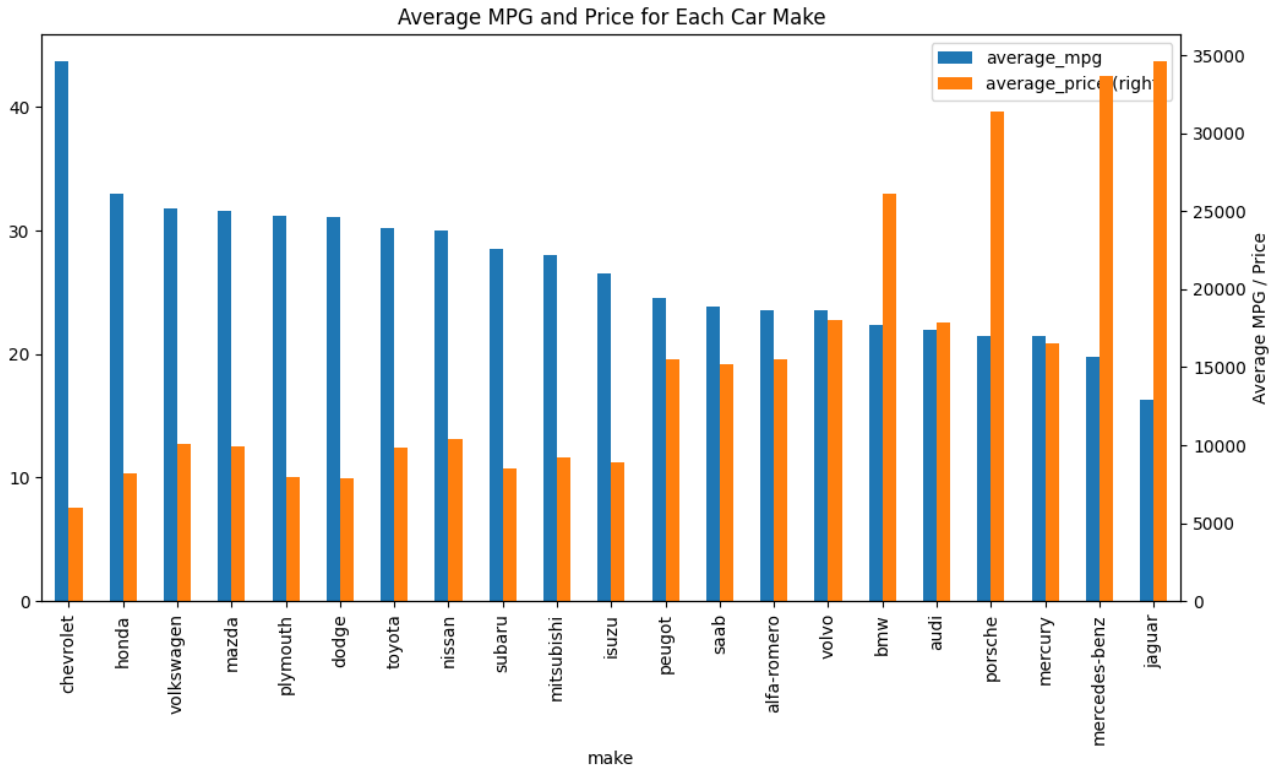
The graph above shows us the average price of each car manufacturer's cars. Despite not having any cars within the top 5 most expensive cars, Jaguar has the highest average price of cars. They are shortly followed by Mercedes-Benz and Porsche. Chevrolet makes the cheapest cars.

2. Average MPG for Each Car Manufacturer:



The above graph shows us the average miles per gallon (MPG) or fuel efficiency of each manufacturer's cars. Chevrolet convincingly has the worst fuel efficiency, while Jaguar has the best.

Comparison of Average MPG and Price:



In the graph above, we can see graphs 1 and 2 side by side, comparing price and average MPG (fuel efficiency). This graph can be used to see whether the price of a car is worth the amount of money saved on fuel.

Immediately, we can see that the worst fuel efficiency is Chevrolet, although one would save the upfront cost as it is the cheapest car manufacturer. On the other hand, Jaguar is the most expensive car manufacturer, yet it has the best fuel efficiency.

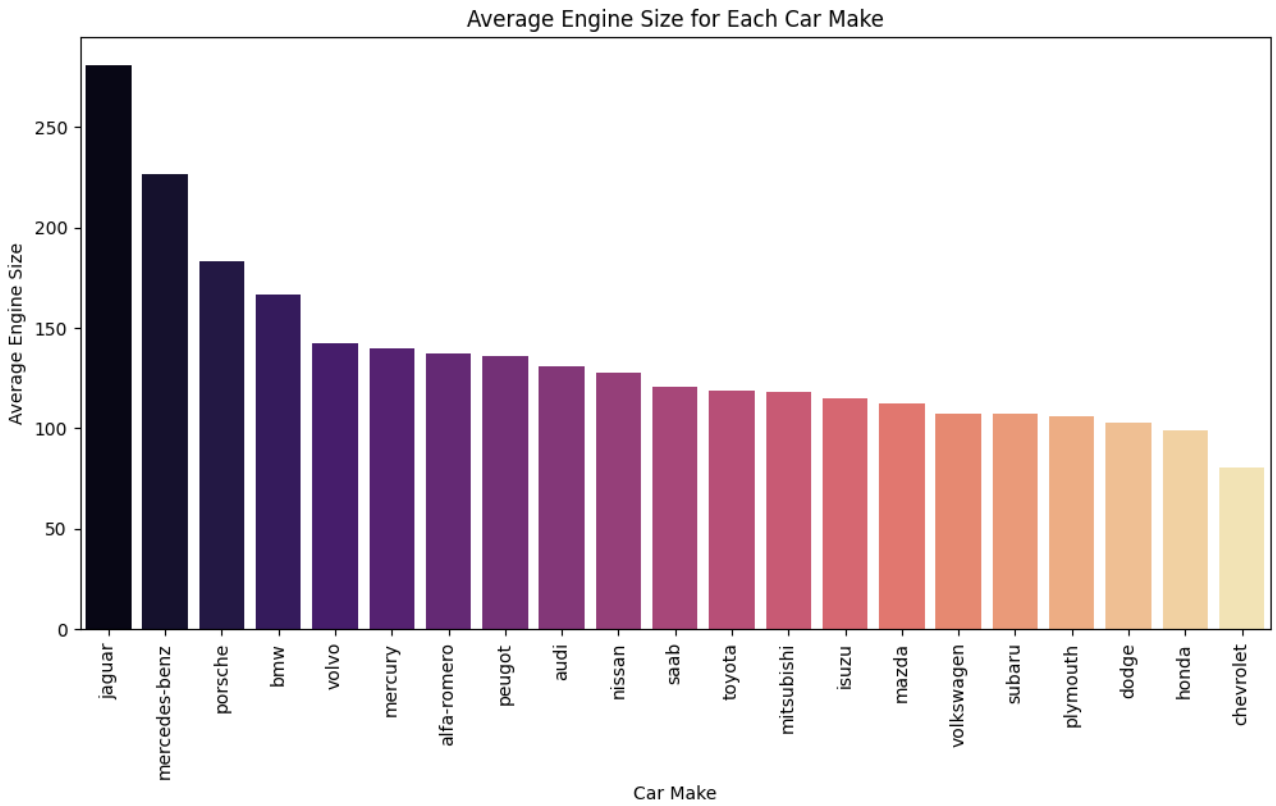
The car manufacturers with the best MPG-to-Price ratio would be Mercury, Audi, Volvo, Alfa Romero, Saab, and Peugeot. These cars won't break the bank and will have a relatively good fuel efficiency.

3. Cars with the Largest Engine Capacity:

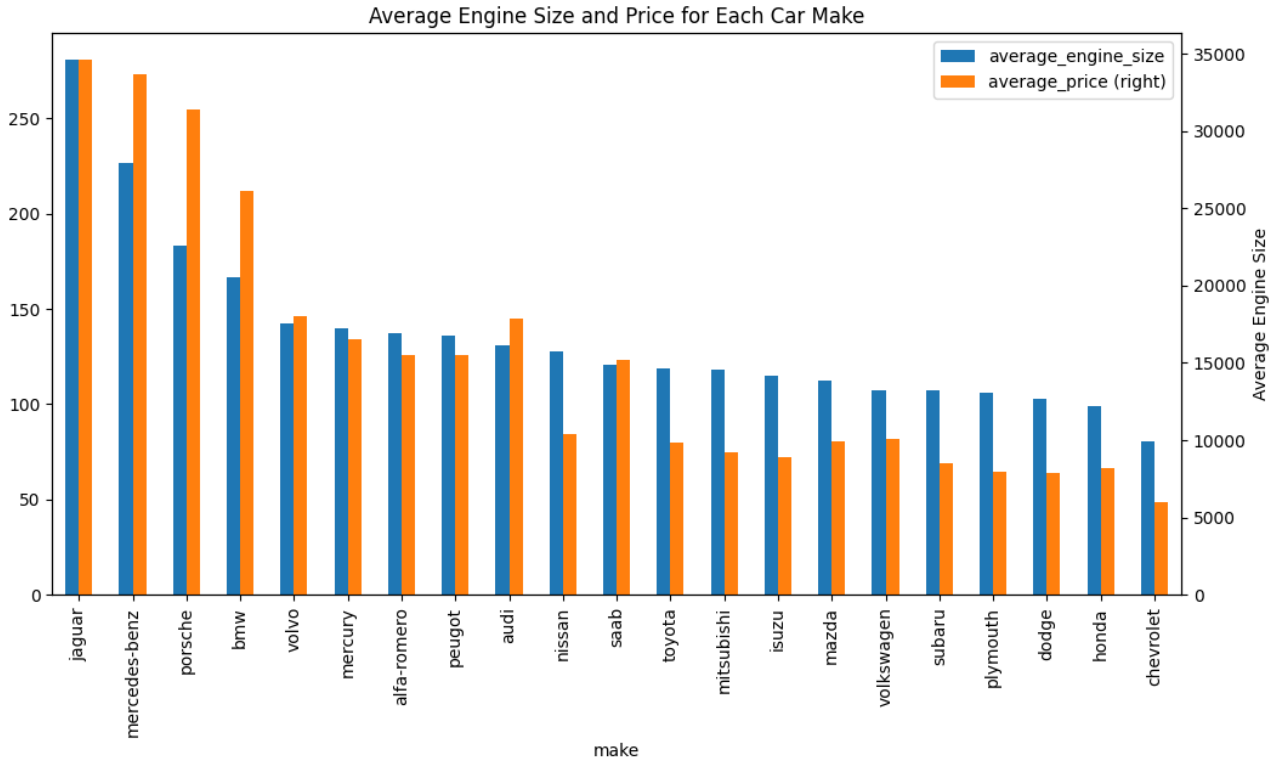
The Top 5 Cars with the Largest Engine Capacity is:
1. #49 Jaguar - 326
2. #73 Mercedes-Benz - 308
3. #74 Mercedes-Benz - 304
4. #48 Jaguar - 258
5. #47 Jaguar - 258

Within the top 5 of largest engine capacity, Jaguar holds the number 1 spot as well as 3 and 4, while Mercedes-Benz holds second and third.

Below is the average engine capacity of each car manufacturer:



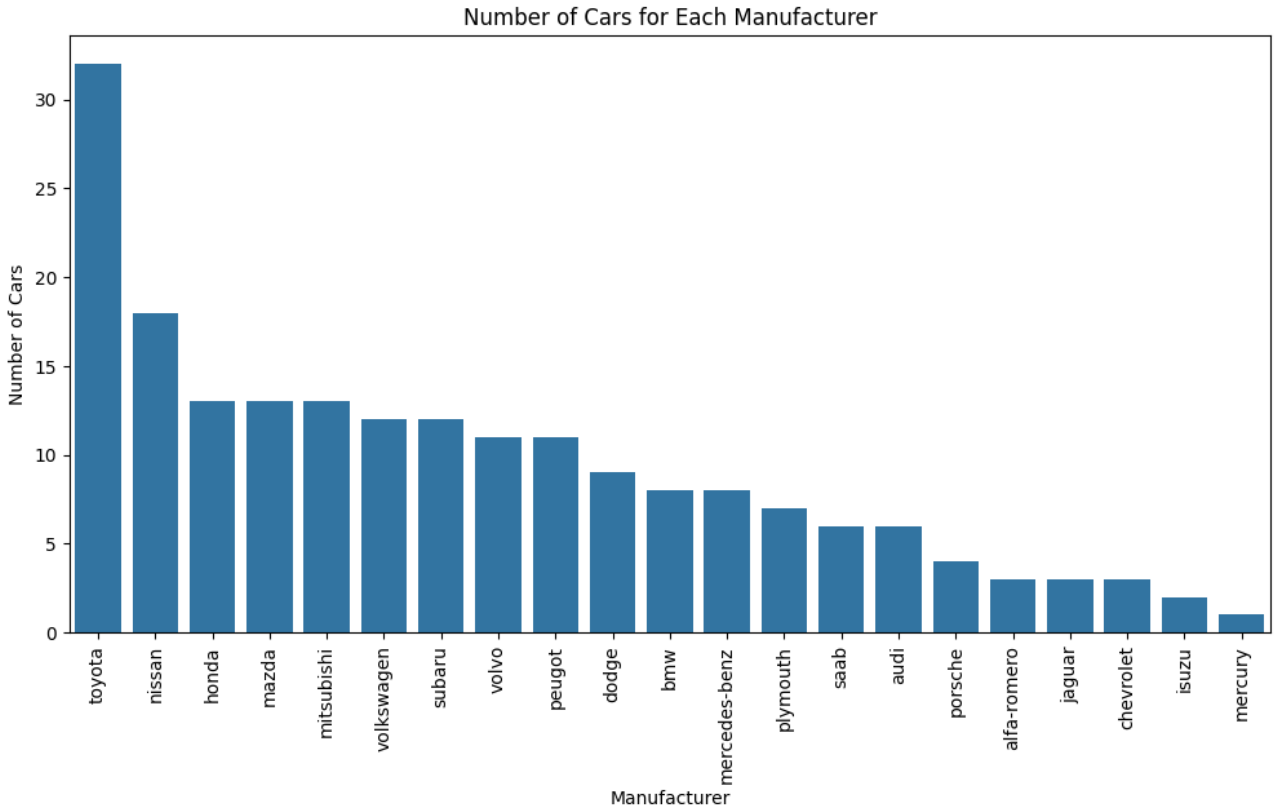
Comparison of Average Engine Size and Price:



In the above graph, we compare the average engine size to price. There is a clear trend that the more expensive the car is, the larger the size of the engine, although the cheaper cars are where you'll get the most out of your money in terms of engine size.

Once again, Mercury, Audi, Volvo, Alfa Romero, Saab, and Peugeot have the most even ratio. With these cars, you will get what you are paying for. From my interpretation, these 6 car brands have the best price, from engine size to fuel efficiency ratio and therefore will be the best "bang for your buck".

4. Vehicle Manufacturers with the most Models:



In the above graph, we can see the manufacturers with the most cars in the dataset. Toyota is a clear winner with 32 cars, followed by Nissan, and then a tied third between Honda, Mazda and Mitsubishi.

It is worth noting that Mercury only has a single car in the dataset, meaning that all of their statistics are based on their one car.

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